

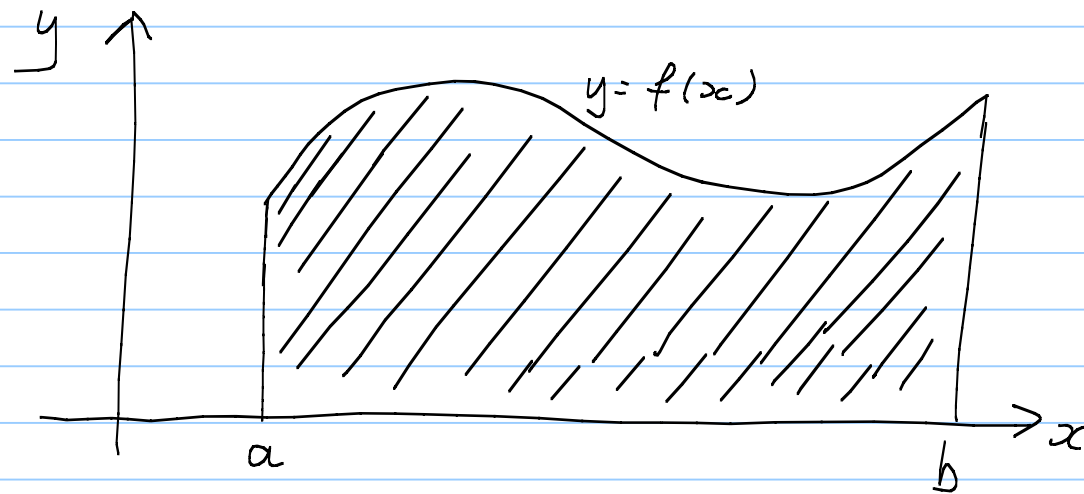
Unit 5.2 Areas and Distances

Note Title

2013/05/12

Section 5.1 in Stewart.

We want to find the area under a curve.



or the distance traveled by a car.

If the function f gives the velocity

(which is a constant function), then the area under the graph gives the distance travelled.

The General problem

If f is continuous on $[a, b]$ and $f(x) > 0$ on $[a, b]$ then we can estimate the area between the graph and the x -axis and the lines $x=a$ and $x=b$.

We will use the sum of areas of rectangles, called a Riemann sum.

We use rectangles with a constant width.

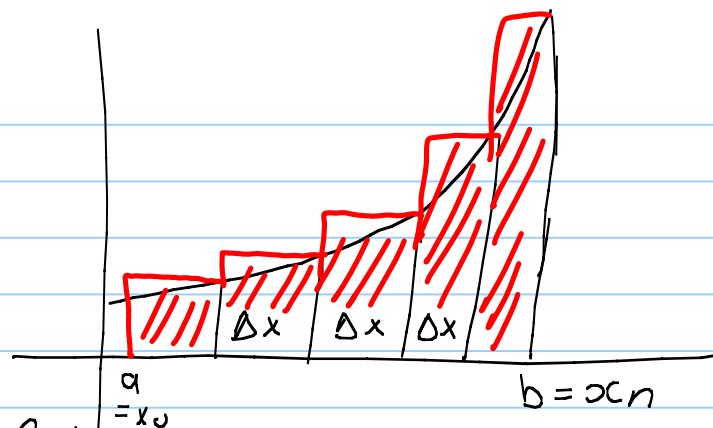
If the interval is $[a, b]$ and we want n rectangles, then the width = $\Delta x = \frac{b-a}{n}$

There are four possibilities to find the height of the rectangle:

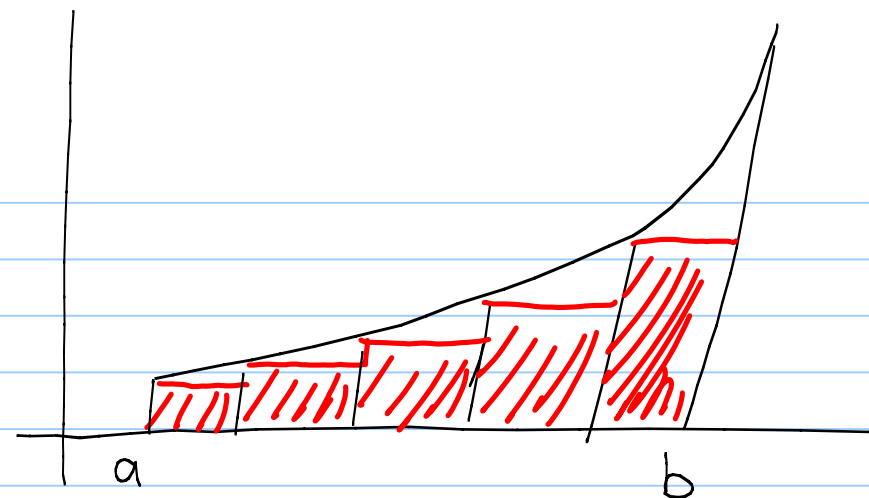
- 1) The left-hand-sum (using the left endpoint of each rectangle)
- 2) The right-hand-sum (using the right endpoint of each rectangle)
- 3) The overestimate (using the maximum function value in each rectangle)

4) The underestimate (using the minimum function value in each rectangle)

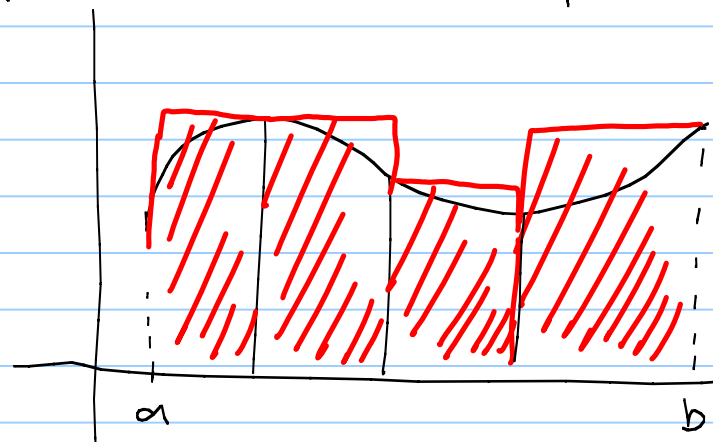
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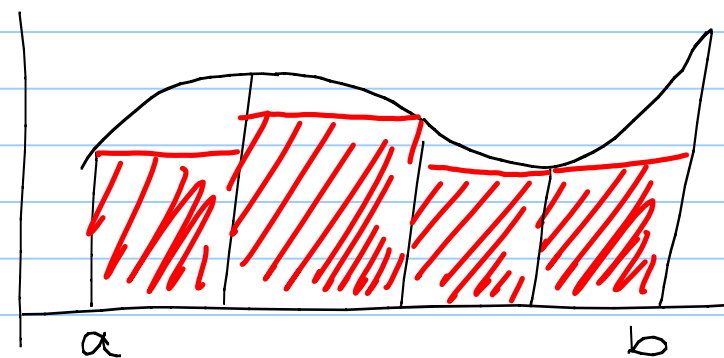
left-hand-endpoints



right-hand-endpoints

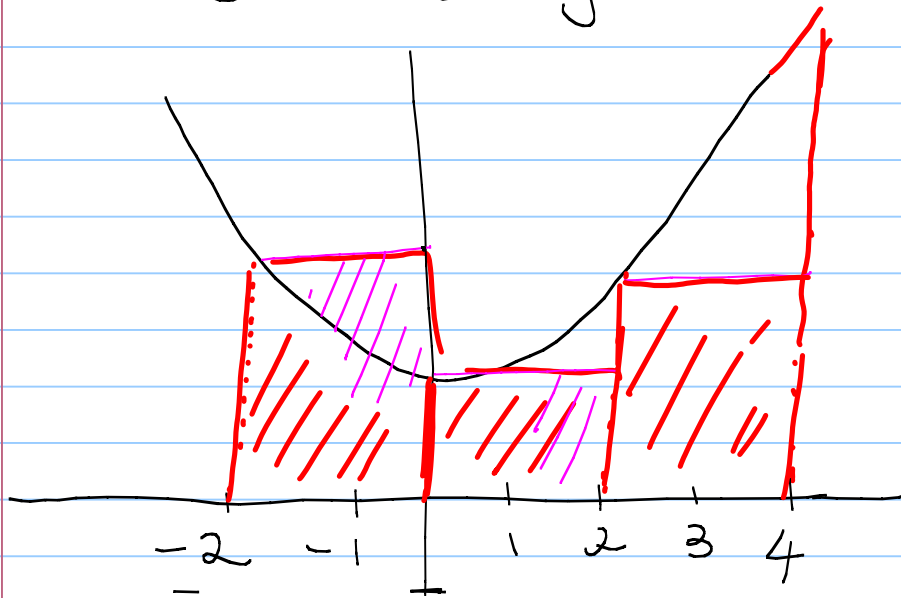


over estimate



underestimate

Example Use rectangles (Riemann sum) to find area under $f(x) = x^2 + 4$ on $[-2, 4]$ using the left-hand-sum and 3 rectangles.

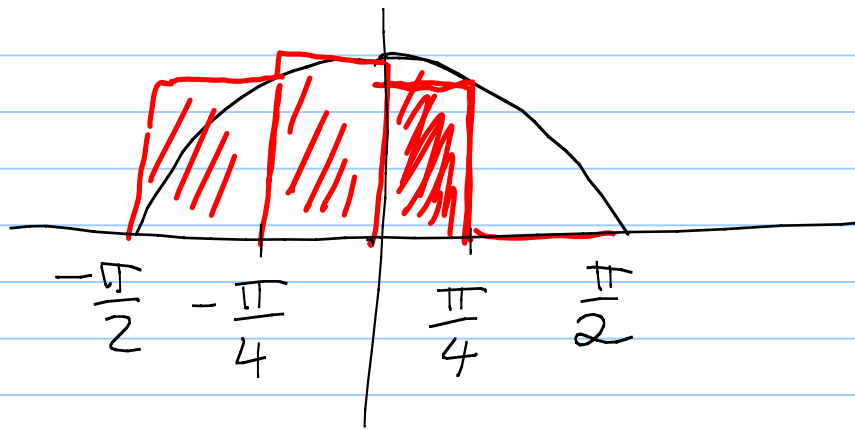


$$\begin{aligned}\Delta x &= \frac{b-a}{n} \\ &= \frac{4 - (-2)}{3} = \frac{6}{3} \\ &= 2\end{aligned}$$

$$\begin{aligned}
 \text{Area} &= f(-2)\Delta x + f(0)\Delta x + f(2)\Delta x \\
 &= 8(2) + 4(2) + 8(2) \\
 &= 40
 \end{aligned}
 \left. \vphantom{\begin{aligned} \text{Area} &= f(-2)\Delta x + f(0)\Delta x + f(2)\Delta x \\ &= 8(2) + 4(2) + 8(2) \\ &= 40 \end{aligned}} \right\} \begin{array}{l} \text{left} \\ \text{-hand} \\ \text{-Sum} \end{array}$$

$$\begin{aligned}
 \text{Area} &= 2f(0) + 2f(2) + 2f(4) \\
 &= 2(4) + 2(8) + 2(20) \\
 &= 8 + 16 + 40 \\
 &= 64
 \end{aligned}
 \left. \vphantom{\begin{aligned} \text{Area} &= 2f(0) + 2f(2) + 2f(4) \\ &= 2(4) + 2(8) + 2(20) \\ &= 8 + 16 + 40 \\ &= 64 \end{aligned}} \right\} \begin{array}{l} \text{Right-Hand} \\ \text{-Sum.} \end{array}$$

Example Use rectangles (Riemann sum) to find area under $f(x) = \cos x$ on $[-\frac{\pi}{2}, \frac{\pi}{2}]$ using the right-hand-sum and using 4 rectangles.



$$\Delta x = \frac{\frac{\pi}{2} - (-\frac{\pi}{2})}{4}$$
$$= \pi/4$$

$$\text{Area} = f(-\pi/4) \Delta x + f(0) \Delta x + f(\pi/4) \Delta x + f(\pi/2) \Delta x$$

$$= \frac{1}{\sqrt{2}} (\pi/4) + 1 (\pi/4) + \frac{1}{\sqrt{2}} (\pi/4) + 0 (\pi/4)$$

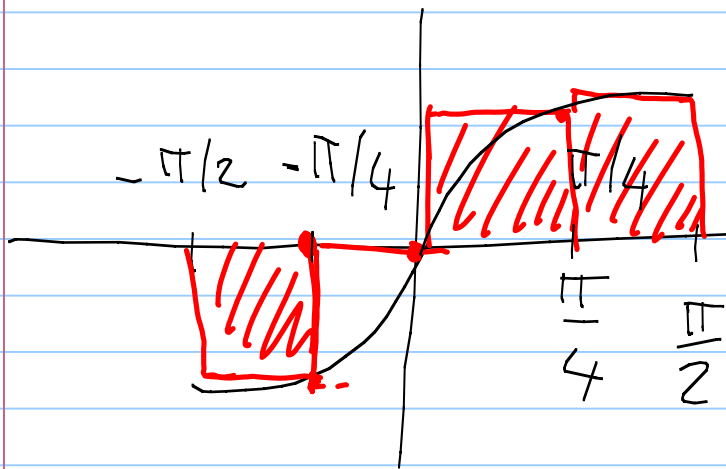
$$= 1,896$$

If the function is above the x -axis we get a positive area. If the function is below the x -axis we get a negative area.

Example The right-hand-sum

- $f(x) = \sin x$ on $[-\frac{\pi}{2}, \frac{\pi}{2}]$ using
4 rectangles.

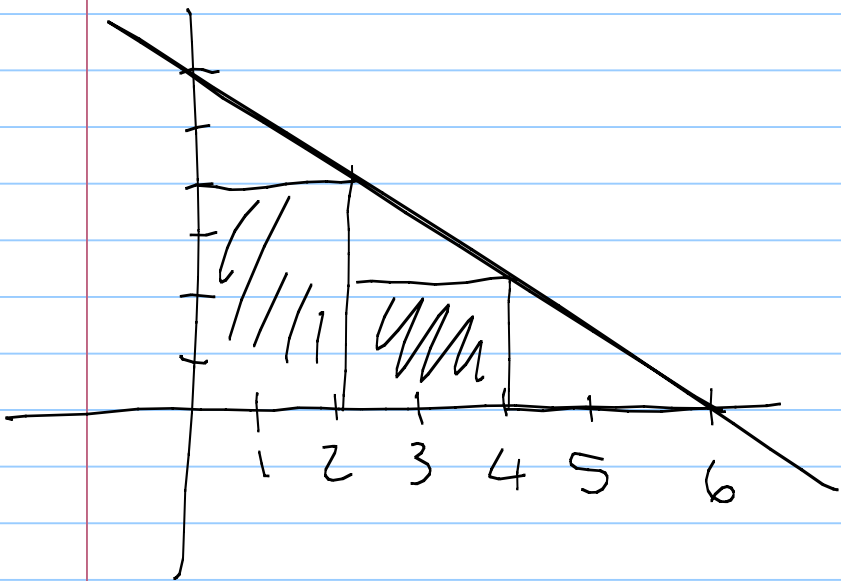
$$\Delta x = \frac{b-a}{4} = \pi/4$$



$$\begin{aligned} \text{Area} &= \pi/4 f(-\pi/4) + \pi/4 f(0) + \pi/4 f(\pi/4) \\ &\quad + \pi/4 f(\pi/2) \\ &= \pi/4 \left(-\frac{\sqrt{2}}{2}\right) + \pi/4 (0) + \pi/4 \left(\frac{\sqrt{2}}{2}\right) + \pi/4 (1) \\ &= \pi/4. \end{aligned}$$

Example 3 Find area under

$f(x) = 6 - x$ on $[0, 6]$ using a
underestimate and 3 rectangles



$$\Delta x = 6/3 = 2$$

$$\text{Area} = 2 f(2) + 2 f(4) + 2 f(6)$$

$$= 2(4) + 2(2) + 2 f(0)$$

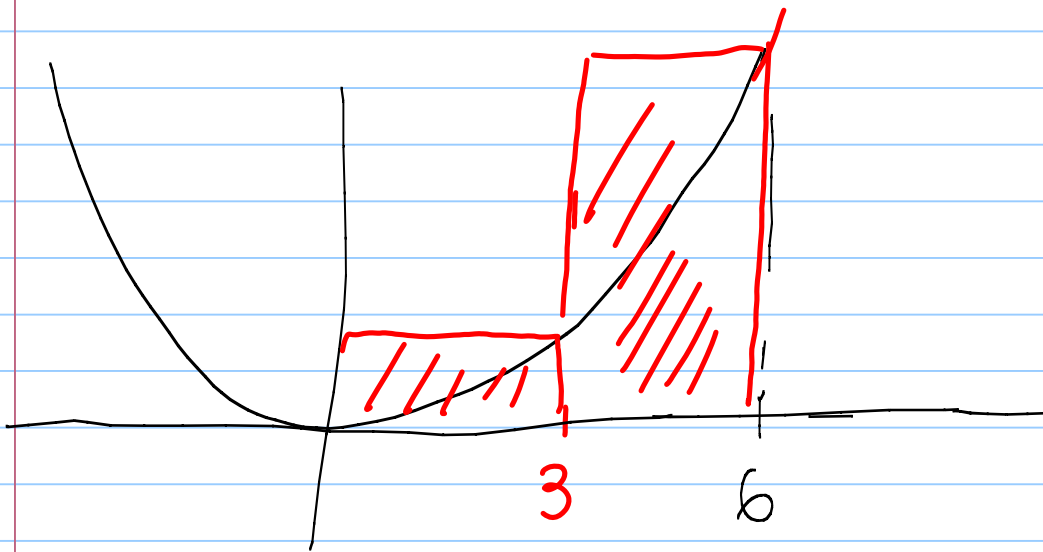
$$= 8 + 4 + 0$$

$$= 12$$

Example

Find area under

$f(x) = x^2$ on $[0, 6]$ using an over-estimate
on $[0, 6]$ and using 2 rectangles.

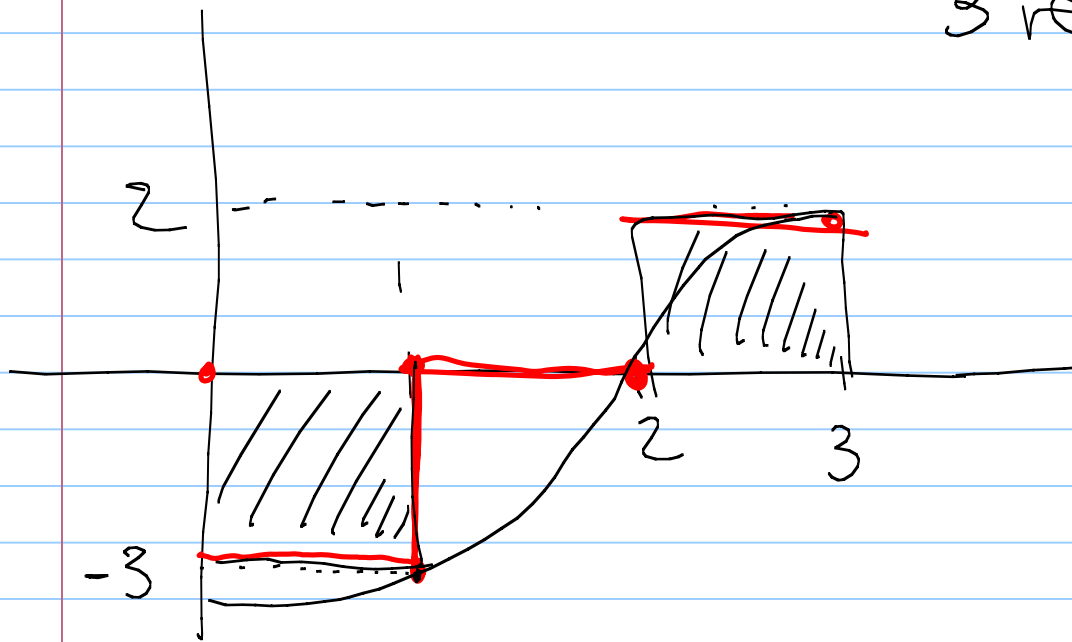


$$\Delta x = \frac{6 - 0}{2} \\ = 3$$

$$\text{Area} = 3 f(3) + 3 f(6) \\ = 135.$$

Example

Find area using
Right-hand-Sum and
3 rectangles



$$\Delta x = \frac{3-0}{3} = 1$$

$$\begin{aligned}\text{Area} &= 1 \times -3 + 1 \times 0 + 1 \times 2 \\ &= -3 + 2 \\ &= -1\end{aligned}$$

